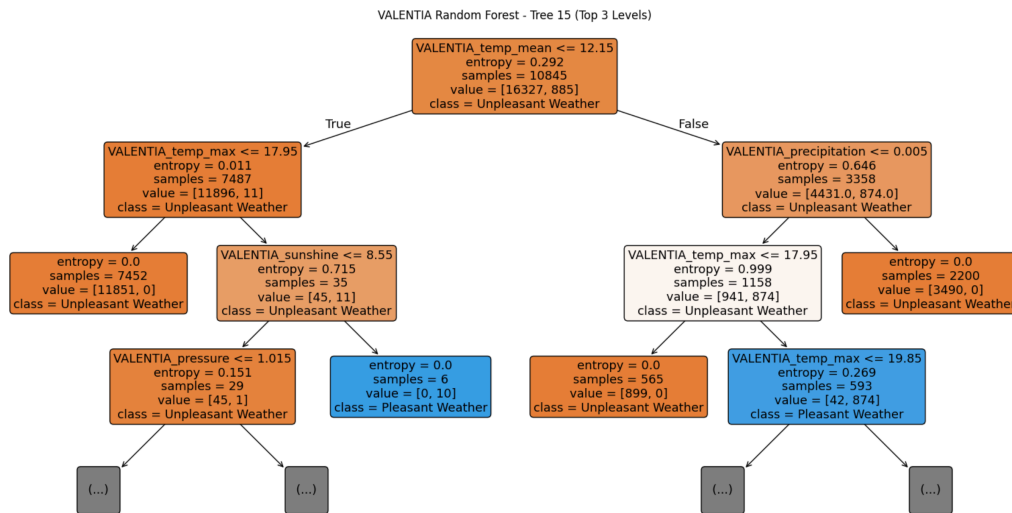


Optimisation

Random Forest 1 – Valentia

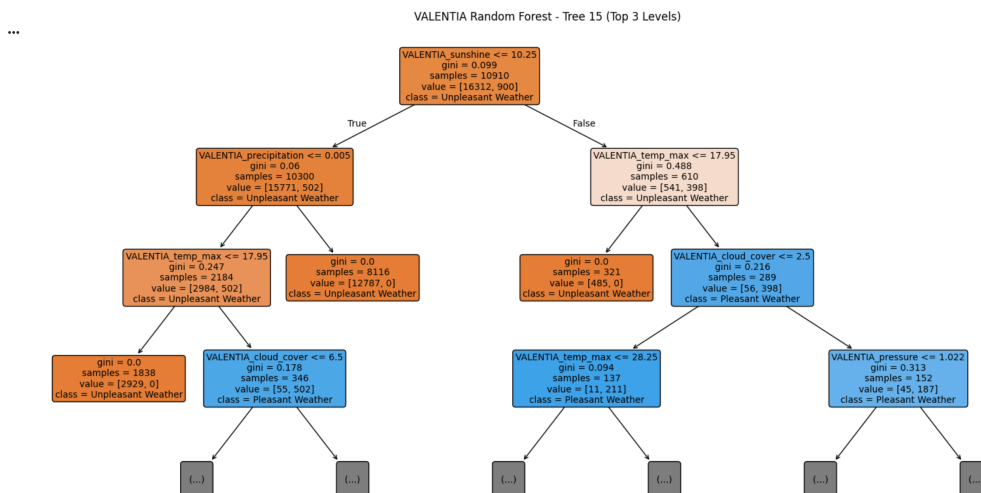
Trialling for 1 weather station (not Sonnblick!). Valentia, with 50 Iterations. Time taken = 4 minutes. Then took 7.1 minutes for 100 iterations. Could estimate would take 4 minutes x 15 stations = at least 1 hour (possibly more with exponential scaling) to run all! Therefore, will need to reduce to a decade when running for all stations.

Optimised – accuracy at 100%

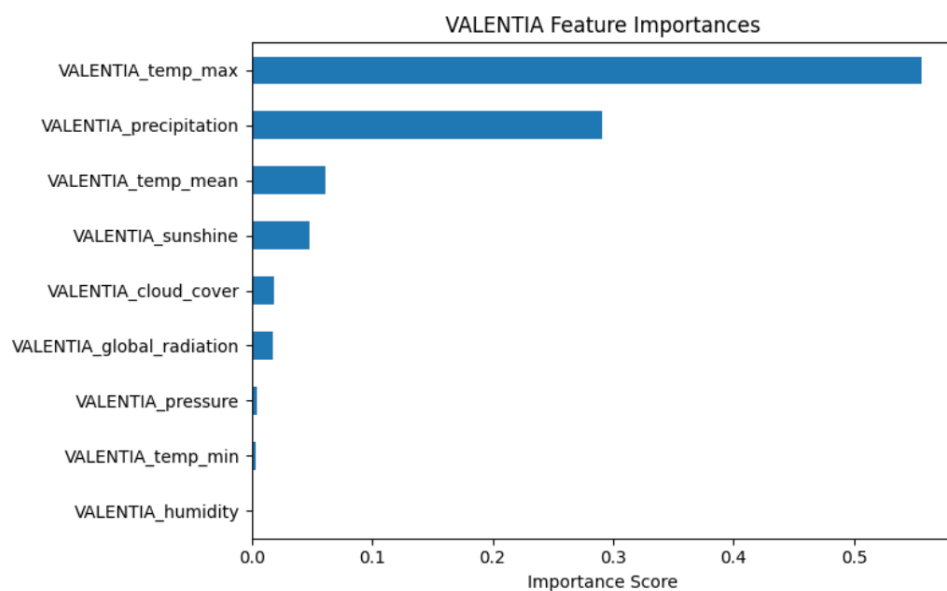


This differs by first splitting on mean temperature being under 12 degrees as unpleasant but also has under around 18 degrees as unpleasant and low rainfall leading onto pleasant weather.

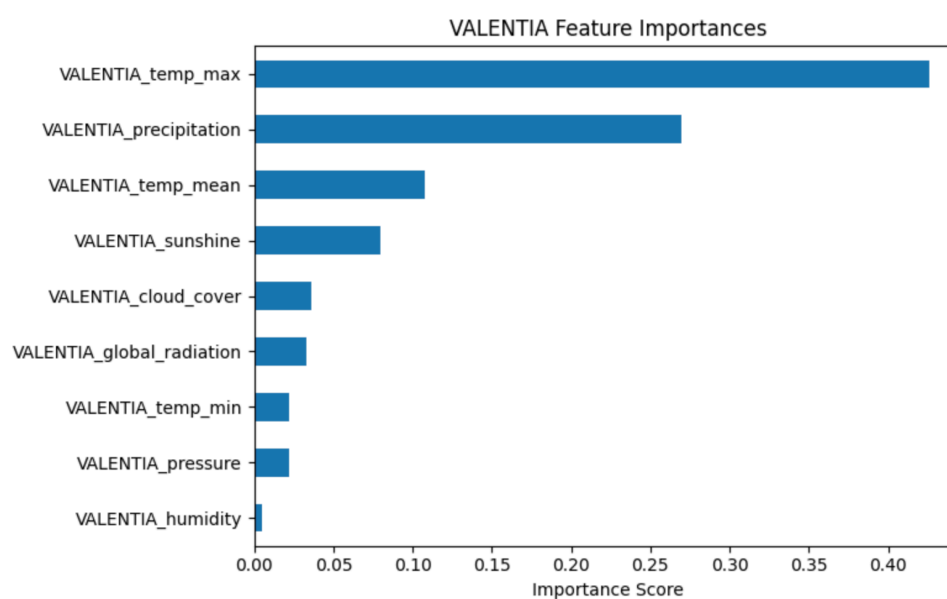
Comparing with non-optimised:



The feature importances at 100 iterations show max temperature being even more important, while the bottom 3 are the same variables but become even less important. This shows with model optimised it is presenting more refined answers while still following the rough pattern so we can be confident of the accuracy.



Comparing with non-optimized:



Random Forest 2 – 2010s

As I couldn't compare with my 2.3 Random Forest which featured all stations. I went back and ran 2.3 for just the 2010s' decade data for a clean comparison.

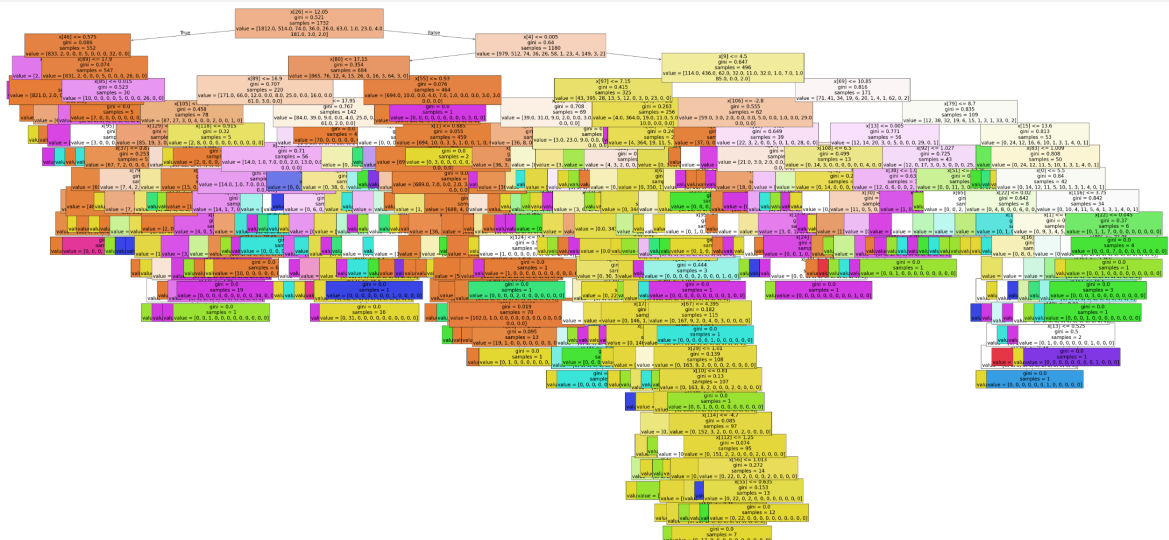
Running for the 2010s. Search took 6 minutes on 10 Iterations and then went for 50 iterations which took 19 minutes but achieved a 96% accuracy on test data and 100% on training data, which is significantly higher than the non-optimized random forest.

2010s – optimized at 96%



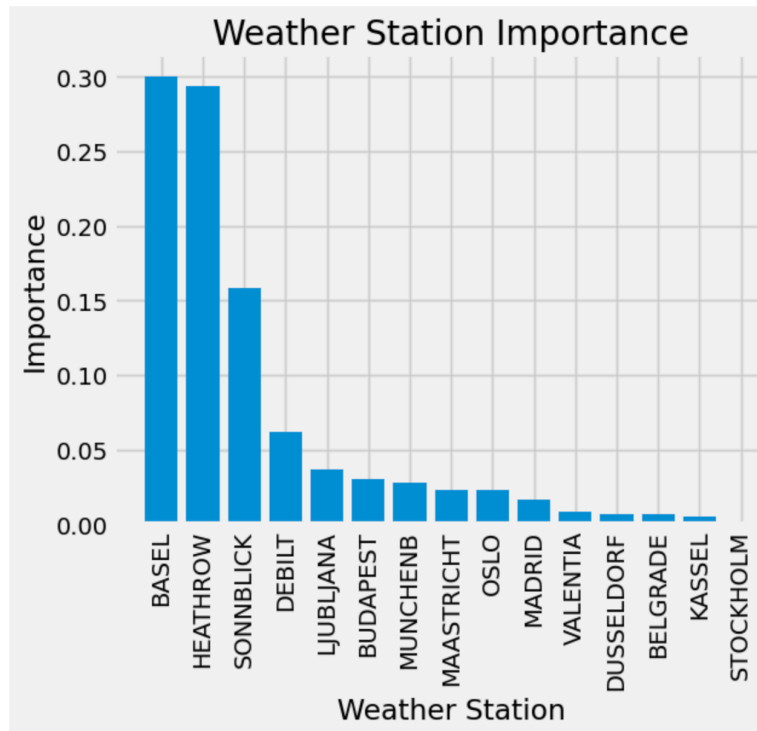
2010s – non-optimized at 89%

```
fig = plt.figure(figsize=(80,40))  
plot_tree(clf.estimators_[15], fontsize = 20, filled=True);
```

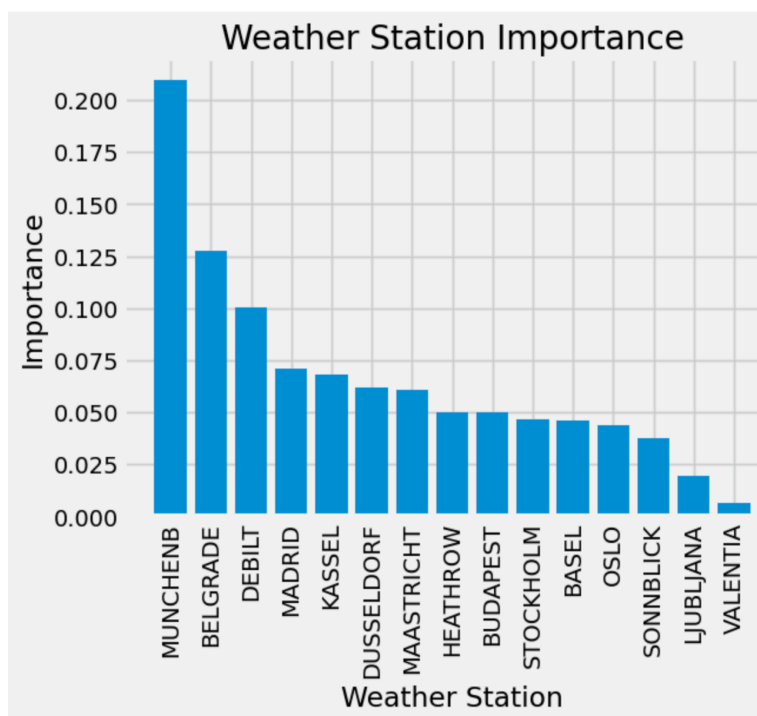


The feature importances significantly change with this increased accuracy. Basel, Heathrow and Sonnblick become the most important, though we know of the issue with the Sonnblick data and so would treat Debilt as the next most important. This shows the importance of using hyperparameters, as with higher accuracy we can trust the results and focus more on these three stations for further analysis, particularly Basel an Heathrow.

2010s – optimised at 96%



2010s – non-optimised at 89%



Part 2 – Deep Learning

I had also investigated the CNN along with RNN back in 2.2, so wanted to try comparing the CNN as should be more appropriate for this data set. However, I encountered some challenges getting the various libraries and packages to integrate, and with so many lines of code there may be other incompatibilities which Gemini AI may not have fully corrected, therefore am not confident the results are optimum.

Accuracy 68%, loss 0.86

This is recognising 7 stations, compared to 2 from non-optimised. While the accuracy is only slightly higher, the loss is converging which seems to be the reason for producing more accurate results. I still think with further refinement the hyperparameters can recognise more of the stations, and it still defaulting to Basel is likely the main accuracy, but is promising to see this improvement.

```
print(confusion_matrix(y_test, y_pred, stations))
```

Pred	BASEL	BELGRADE	BUDAPEST	DEBILT	HEATHROW	LJUBLJANA	MADRID
True							
BASEL	3412	214	0	22	3	1	30
BELGRADE	580	480	0	5	0	5	22
BUDAPEST	141	52	0	4	3	6	8
DEBILT	64	3	0	15	0	0	0
DUSSELDORF	22	1	0	6	0	0	0
HEATHROW	52	5	0	7	10	2	6
KASSEL	9	1	0	0	1	0	0
LJUBLJANA	35	12	0	1	0	9	4
MAASTRICHT	9	0	0	0	0	0	0
MADRID	199	51	1	8	2	5	192
MUNCHENB	6	1	0	0	0	0	1
OSLO	3	1	0	1	0	0	0
STOCKHOLM	3	1	0	0	0	0	0
VALENTIA	1	0	0	0	0	0	0

Compared to non-optimised:

Accuracy: 64%, loss 121068952

```
print(confusion_matrix(y_test, model.predict(X_test)))
```

180/180			1s	3ms/step
Pred	BASEL	VALENTIA		
True				
BASEL	3680	2		
BELGRADE	1092	0		
BUDAPEST	214	0		
DEBILT	82	0		
DUSSELDORF	29	0		
HEATHROW	82	0		
KASSEL	11	0		
LJUBLJANA	61	0		
MAASTRICHT	9	0		
MADRID	458	0		
MUNCHENB	8	0		
OSLO	5	0		
STOCKHOLM	4	0		
VALENTIA	1	0		

Part 3 – Iteration

- Which model would you use for each iteration? Expand on your observations from the random forest and deep learning models.

I would break down the information down into decades, to compare the station importances over time. Then could analyse which stations frequently are highest and then merit independent analysis. I would prefer to use Random Forest as achieved high accuracy, and for Valentia, the feature importances were similarly consistent but became more refined.

- What variables would you recommend that Air Ambulance pay the most attention to while deciding whether it's safe to fly?

Severe weather conditions such as wind, rain, snow, along with extreme weather which could include max temperatures. This is currently limited as we know the data is partial for wind (11 stations) and especially snow, (7 stations) so would need to retrieve fuller data. But if we focus on Heathrow and Basel and Debilt which are the highest for importance from our Random Forest in 2010s, they have data for all conditions (Heathrow and Debilt just missing Snow) so we can at least perform refined local analysis for dangerous conditions.